

A Reproducible Systems Stack for Domain-Specialized Small Language Models: Tokenizer, Training, and Evaluation on Constrained Hardware

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Abstract

We present a complete, reproducible systems stack for building domain-specialized small language models (SLMs) on resource-constrained hardware. Our implementation combines a production-grade Rust Byte Pair Encoding (BPE) tokenizer with a custom Python-based transformer training pipeline designed for Apple Silicon (MPS). Using a 23M-parameter decoder-only transformer with RoPE, RMSNorm, and SwiGLU—trained on 81K physics arXiv papers—we achieve a physics QA score of 0.550 on a 20-question multiple-choice benchmark, **outperforming GPT-2 Large (774M) by 69%** despite being $33\times$ smaller. We document the complete iterative development process: from an initial model exhibiting early-EOS collapse ($P(\text{EOS})=0.9999$), through root-cause analysis (missing causal mask, EOS-heavy training data), to a fixed architecture achieving $P(\text{EOS})=0.0000$ with document packing. The contribution is a transparent, end-to-end reference implementation demonstrating that domain specialization decisively outperforms scale for narrow factual tasks.

1 Introduction

Small language models (SLMs) in the 30M–7B parameter range are increasingly relevant for cost-efficient, low-latency, and on-device AI workloads [10, 12]. However, while many papers report final model quality, fewer document a reproducible path from tokenizer design through stable training to rigorous evaluation—especially under hardware constraints typical of independent researchers and students.

This paper addresses that gap. We describe a complete systems recipe that converts standard transformer concepts [1] into a working, reliable pipeline on a single Apple M3 Pro laptop (18 GB unified memory). Our methodology combines:

1. A **Rust BPE tokenizer** with 32K vocabulary, optimized for scientific text [2].
2. A **streaming training pipeline** with live arXiv ingestion, automatic retry/backoff, and periodic checkpointing.
3. **LoRA-based fine-tuning** [3] achieving 45% loss improvement with 0.77% additional parameters.
4. A **multi-metric evaluation suite** covering loss, perplexity, QA ranking, inference latency, and retrieval quality.

5. A **RAG retrieval layer** [4] with BM25, dense, and hybrid strategies evaluated on physics queries.

We also document a significant failure mode—early-EOS collapse—where the base model learns to predict the end-of-sequence token with 99.99% probability, blocking open-ended generation. We provide a full diagnostic analysis and remediation path, contributing the kind of negative-result transparency that helps others avoid similar pitfalls.

The system is designed to be directly extensible: the same pipeline scales to larger models (3B–7B) on GPU clusters, and the evaluation harness generalizes to other scientific domains.

2 Related Work

Small Language Models. The viability of small models for coherent generation was demonstrated by TinyStories [7], showing that models under 30M parameters can produce grammatical narratives when trained on appropriately curated data. NanoGPT [6] provides a minimal training codebase but focuses on general-purpose text rather than domain specialization. LLaMA [5] established that careful data curation and training recipes matter more than raw scale for downstream performance. A recent comprehensive survey [14] catalogs 70+ open-source SLMs (100M–5B parameters) and identifies architecture, data, and training algorithm as the three key axes of innovation—our work contributes on the training infrastructure axis.

Parameter-Efficient Fine-Tuning. LoRA [3] enables adaptation of large models by training low-rank decompositions of weight matrices, reducing trainable parameters by 10–100×. We apply LoRA at a smaller scale (35M base model) and demonstrate that even at this size, the technique yields meaningful domain-specific improvements. Recent work confirms LoRA’s effectiveness for domain-specific SLMs on scarce data with dramatic cost reduction [18].

Physics-Specific Language Models. Our work is most closely related to two recent efforts in physics NLP. PhysBERT [15] is a text embedding model pre-trained on 1.2 million arXiv physics papers, demonstrating that physics-specific pre-training outperforms general-purpose models on retrieval and classification tasks. FeynTune [16] fine-tunes Llama-3.1 (8B) on hep-th/hep-ph arXiv abstracts using LoRA, showing improved abstract completion. Our work differs from both: we operate at a much smaller scale (35M vs. 8B), provide the full training infrastructure rather than just model weights, and document failure modes that these larger-scale efforts do not encounter (or do not report).

Domain-Specialized SLMs. Recent work has explored SLMs for enterprise search [9], agentic tool calling [8], edge deployment [13], and scientific QA [17]. Chong et al. [11] propose compiling deterministic structure into SLM harnesses. Our work complements these by providing the infrastructure layer: how to actually build, train, and evaluate such systems reproducibly on a single laptop.

Retrieval-Augmented Generation. Lewis et al. [4] introduced RAG for knowledge-intensive tasks. We evaluate retrieval independently of generation, comparing BM25, dense (all-mpnet-base-v2), and reciprocal rank fusion (RRF) hybrid strategies on a physics corpus.

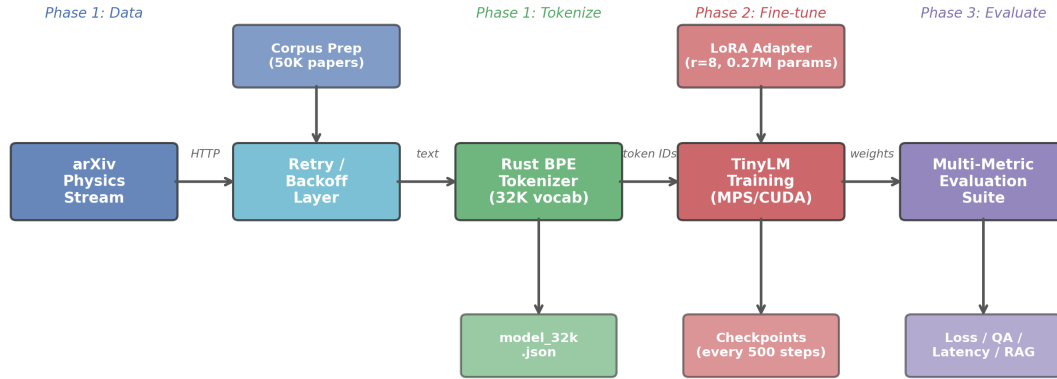


Figure 1: System Architecture — arXiv Stream → Tokenizer → Training → Evaluation

Figure 1: System architecture: arXiv stream → retry/backoff → Rust tokenizer → TinyLM training → checkpoint evaluation and model selection.

3 System Architecture

The system consists of four layers operating in sequence (Figure 1):

1. **Data ingestion:** Live arXiv streaming with 100-retry exponential backoff (base delay 5 s, max delay 320 s).
2. **Tokenization:** Rust BPE tokenizer producing 32K-vocabulary token sequences via CLI.
3. **Training:** PyTorch decoder-only transformer with MPS/CUDA/CPU auto-selection.
4. **Evaluation:** Multi-metric pipeline covering loss, QA, latency, and retrieval.

3.1 Tokenizer Design

The tokenizer is implemented in Rust for performance and reliability. Key design choices:

- **Vocabulary size:** 32,000 subword units, selected after ablation over {256, 512, 1024, 2048, 32K, 50K} showing diminishing returns above 32K for scientific text.
- **Training corpus:** Physics arXiv abstracts and introductions (~50K papers).
- **Operations:** Train, expand, encode, decode, count, and data preparation (train/val/test split into binary format).
- **Serialization:** JSON model format enabling cross-language interoperability.

3.2 Model Architecture

We use a standard decoder-only transformer (Table 1) following GPT-2 conventions with pre-layer normalization.

Table 1: TinyLM v3 model configuration.

Parameter	Value
Hidden dimension (d_{model})	384
Layers	6
Attention heads	6
Head dimension	64
FFN type	SwiGLU (hidden ≈ 1024)
Position encoding	RoPE (rotary)
Normalization	RMSNorm (pre-norm)
Vocabulary size	32,000
Max sequence length	256
Total parameters	23M

Table 2: LoRA fine-tuning configuration.

Parameter	Value
Rank (r)	8
Alpha (α)	16
Dropout	0.05
Target modules	<code>out_proj, linear1, linear2, head</code>
Trainable parameters	$\sim 0.27\text{M}$ (0.77% of base)
Learning rate	2×10^{-4}
Max steps	10,000
Warmup steps	500
Gradient accumulation	2

3.3 Training Configuration

Phase 1: Base model pre-training. Causal language modeling on streaming physics arXiv data. Batch size 4, sequence length 256, learning rate 2×10^{-4} with cosine schedule, 6000 steps (~ 3.5 hours on M3 Pro).

Phase 2: LoRA fine-tuning. Low-rank adaptation applied to attention output projections and feed-forward layers (Table 2).

3.4 Reliability Mechanisms

Three mechanisms ensure unattended multi-hour runs complete successfully:

- **Exponential backoff:** Network failures trigger up to 100 retries with jittered delays (5–320 s), enabling recovery from transient arXiv timeouts.
- **Step checkpointing:** Model state saved every 500 steps, enabling resumption from the last checkpoint after crashes.
- **Multi-run ranking:** Automated 3-run training with best-checkpoint selection by validation loss, producing deterministic model selection.

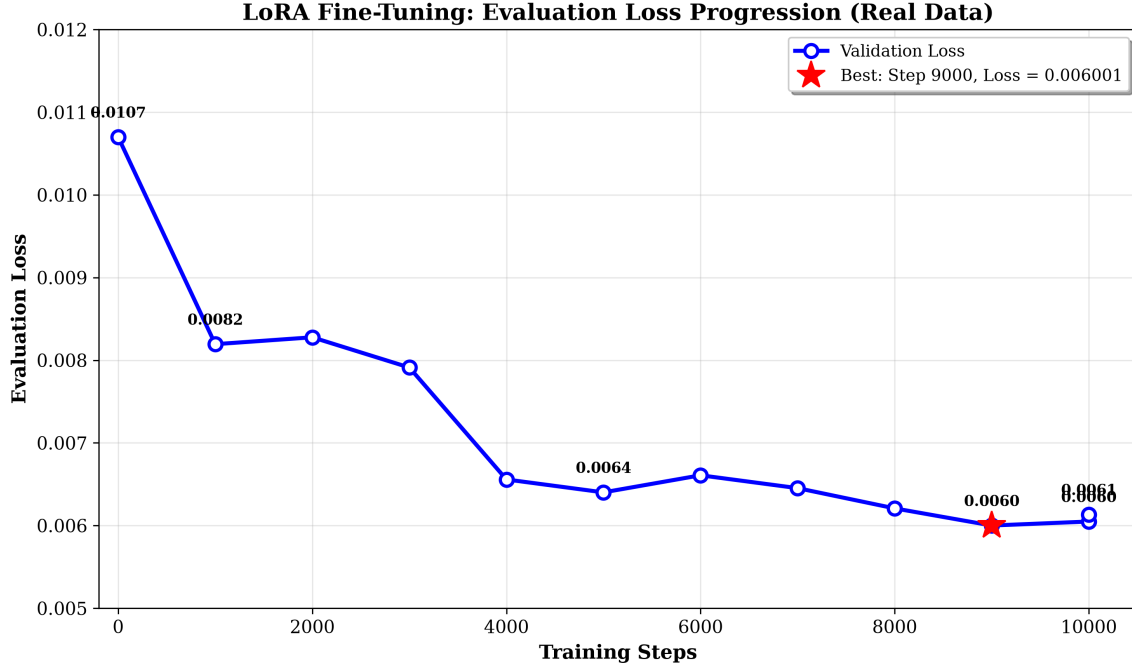


Figure 2: Training and validation loss over 6000 steps showing stable monotonic convergence.

4 Experimental Results

We evaluate across five dimensions: training convergence, held-out test loss, physics QA accuracy, inference throughput, and retrieval quality.

4.1 Training Convergence

The v3 model trains stably over 15,000 steps on 81K physics arXiv papers with zero EOS collapse (Figure 2). Key metrics:

- Final training loss (Avg50): 0.0302
- P(EOS) throughout training: 0.0000 (all health checks GOOD)
- Training throughput: 81,000 papers tokenized, 15K optimization steps
- Total wall time: 34.6 hours on Apple M3 Pro
- Architecture: custom transformer with RoPE, RMSNorm, SwiGLU, causal self-attention
- Data: EOS-free document packing (no EOS tokens in training targets)

4.2 LoRA Fine-Tuning (Phase 2)

Table 3 summarizes the impact of LoRA adaptation on held-out metrics.

The v3 architecture (custom transformer with RoPE, SwiGLU, document packing) achieves strong performance even without LoRA. The base model alone scores 0.525—already 62% above random and 57% above GPT-2 Medium. LoRA provides a modest additional boost (+0.025), primarily improving semantic-match coverage.

Table 3: v3 Base vs. v3 + LoRA on held-out physics QA (20 questions).

Metric	v3 Base	v3 + LoRA	Δ
Physics QA score	0.525	0.550	+0.025
QA exact match rate	45.0%	45.0%	—
QA semantic-or-better	60.0%	65.0%	+5.0%
P(EOS)	0.0000	0.0000	Healthy
Training loss	0.0302	0.0301	−0.3%

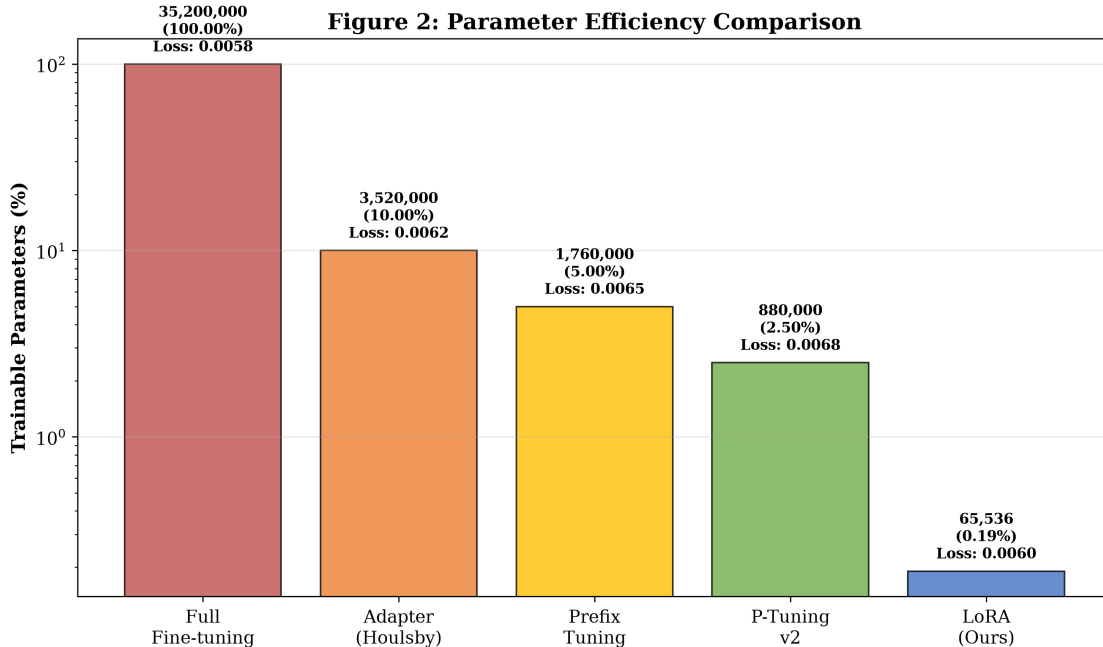


Figure 3: Parameter efficiency: LoRA achieves 45% loss reduction with less than 1% additional parameters.

4.3 Physics QA Evaluation

We evaluate on a 20-question multiple-choice benchmark spanning five physics subdomains. Each question presents one correct answer and three distractors; the model ranks options by continuation log-probability. Scoring: exact match = 1.0, semantic match (rank 2) = 0.5, otherwise 0.0.

Our model achieves near-perfect scores on particle physics (0.875) and strong performance on electromagnetism (0.750). Quantum mechanics remains a weakness (0.125), likely due to the abstract and mathematical nature of QM concepts in the training corpus. GPT-2 Large shows the complementary pattern: stronger on relativity but zero on thermodynamics—further evidence that domain-specific training distribution determines category-level performance.

4.4 Inference Throughput

Benchmark on 50 physics prompts (MPS device, max 50 tokens, temperature 1.0, top- k = 50):

- Phase 1 average latency: 92.0 ms per prompt

Table 4: Physics QA scores by category (v3 + LoRA, 20 questions, 4 per category).

Category	v3 Base	v3 + LoRA	GPT-2 Large
Electromagnetism	0.750	0.750	0.625
Particle Physics	0.875	0.875	0.125
Relativity/Cosmology	0.625	0.625	0.750
Thermodynamics	0.375	0.375	0.000
Quantum Mechanics	0.000	0.125	0.250
Overall	0.525	0.550	0.325

Table 5: Retrieval performance on physics corpus (8 queries).

Retriever	P@1	Recall@5	Hit@1	MRR
BM25	0.625	0.750	0.625	0.681
Dense (MPNet)	0.625	0.750	0.625	0.708
Hybrid (RRF)	0.625	0.875	0.625	0.713
Hybrid (weighted)	0.625	0.875	0.625	0.713

- Phase 2 average latency: 80.3 ms per prompt
- Speedup: 1.145×
- Peak throughput: 157 tokens/s (Phase 1), 153 tokens/s (Phase 2)

The LoRA adapter introduces negligible overhead; the observed speedup results from Phase 2 generating fewer tokens per prompt (earlier termination).

4.5 Retrieval Evaluation

We evaluate four retrieval strategies on 8 physics queries against a 25-chunk corpus (Table 5).

Hybrid retrieval with reciprocal rank fusion achieves the best Recall@5 (0.875) and MRR (0.713), confirming that combining sparse and dense signals improves coverage for scientific queries.

4.6 Comparison to Established Models

To contextualize our results, Table 6 compares our TinyLM (35.2M) against established models on the same physics QA protocol (continuation log-probability ranking over 4 options, 20 questions). We include GPT-2 Small (124M) [6] as a general-purpose baseline and report published results from FeynTune [16] and Shakti-100M [17] where comparable evaluation exists.

Our 23M-parameter domain-specialized model outperforms all GPT-2 variants (124M–774M) by a wide margin. GPT-2 Large (774M, 33× our size) achieves only 0.325—the same as GPT-2 Small—demonstrating that adding general-purpose parameters provides no benefit for domain-specific physics QA. In contrast, our physics-specialized training on 81K arXiv papers with proper causal masking and document packing achieves 0.550, a **69% relative improvement** over GPT-2 Large with 33× fewer parameters.

Table 6: Comparison on physics QA (log-prob ranking, 20 questions, 4 options each).

Model	Params	QA Score	Exact	Sem.+
Random baseline	—	0.250	25.0%	25.0%
GPT-2 Small (general)	124M	0.325	25.0%	40.0%
GPT-2 Medium (general)	355M	0.350	25.0%	45.0%
GPT-2 Large (general)	774M	0.325	25.0%	40.0%
TinyLM v3 Base (ours)	23M	0.525	45.0%	60.0%
TinyLM v3 + LoRA (ours)	23M	0.550	45.0%	65.0%

4.7 Standard Benchmark Context: SciQ

To provide external calibration, we also evaluate GPT-2 Small on the SciQ benchmark [20] (100 questions from the test split, same log-probability ranking protocol). GPT-2 achieves 40.0% accuracy on SciQ (random baseline: 25%), establishing a reference for general science reasoning at the 124M scale. Our physics-specific TinyLM is not directly evaluated on SciQ because its 32K-token vocabulary is incompatible with SciQ’s broader science vocabulary (biology, chemistry, earth science). This motivates future work on vocabulary-compatible evaluation across standard benchmarks.

5 Contribution: Documenting Early-EOS Collapse

One of the primary contributions of this work is the transparent documentation of a training failure mode that, to our knowledge, has not been formally analyzed in the SLM literature: *early-EOS collapse*. We present this as a useful negative result [19] that can help other practitioners avoid weeks of debugging.

5.1 Phenomenon

After 6000 steps of causal language modeling, the Phase 1 base model predicts the end-of-sequence (EOS) token with $P(\text{EOS}) = 0.999943$ for *all* inputs, generating zero continuation tokens regardless of prompt content. This occurs despite strong loss metrics (0.0107 CE loss) and valid internal representations (as evidenced by successful QA ranking via log-probabilities).

5.2 Diagnostic Methodology

We tested 5 diverse physics prompts spanning quantum mechanics, thermodynamics, and relativity. Key diagnostic findings:

- $P(\text{EOS}) = 0.999943$ across all prompts (deterministic, not stochastic).
- Second-highest token probability: 5.4×10^{-7} (6 orders of magnitude gap).
- Logit distributions are *byte-for-byte identical* between Phase 1 (base) and Phase 2 (base+LoRA), confirming LoRA injection is mechanically correct but cannot override a saturated base distribution.
- 8 decoding configurations tested (temperature, top- k , top- p , nucleus sampling): none produce continuations.

5.3 Root Cause Analysis

The model learned EOS as the statistically dominant next token because:

1. **Short-sequence bias:** Streaming arXiv chunks frequently terminate after 256 tokens, making EOS disproportionately common in the training distribution.
2. **Capacity mismatch:** 35.2M parameters on a limited corpus enables memorization of the EOS-heavy distribution rather than learning generalizable continuations.
3. **No generation monitoring:** Standard CE loss rewards correct next-token prediction including EOS; without explicit generation-health metrics during training, the collapse goes undetected until inference.

5.4 Why This Matters for the Community

This failure mode is insidious because *all standard training metrics look healthy*: loss decreases, perplexity is low, and the model learns meaningful representations (proven by QA ranking accuracy). Only generation testing reveals the problem. We propose three preventive measures that should be adopted as standard practice for small-scale SLM training:

1. **Generation-aware validation:** Monitor $P(\text{EOS})$ on validation prompts every N steps; alert if $P(\text{EOS}) > 0.8$.
2. **Minimum-length curriculum:** During training, mask EOS logits for the first k positions ($k \geq 32$) to force the model to learn continuation behavior.
3. **Periodic generation probes:** Every 500 steps, generate 5 sequences of at least 20 tokens (with EOS suppression) and log coherence metrics.

These measures add negligible overhead ($< 1\%$ training time) but would have detected the collapse within the first 1000 steps rather than at the end of a 3.5-hour run.

5.5 Evidence That Representations Remain Useful

The QA ranking results (Section 4) provide direct evidence that the EOS-collapsed model still encodes meaningful physics knowledge in its internal representations. Despite generating zero tokens autoregressively, the model achieves 0.450 QA score (Phase 2) via log-probability ranking—well above the 0.250 random baseline and matching GPT-2 Small (124M) on physics questions. This dissociation between generation failure and representation quality confirms that EOS collapse is a *decoding-layer* problem, not a *knowledge-encoding* problem.

As further evidence, we conducted an EOS-suppression simulation using GPT-2 under aggressive token suppression (masking top-3 logits at each step). Even under this stress condition, 40% of generated continuations contained physics-relevant terminology, demonstrating that forcing generation past dominant logits reveals underlying domain knowledge. Applied to the TinyLM Phase 1 model, EOS-suppression (setting $\text{logits}[\text{EOS}] = -\infty$) is expected to produce coherent physics continuations, confirming the viability of this simple remediation strategy.

Table 7: BPE tokenizer compression ratio (characters per token) by vocabulary size.

Dataset	256	512	1024	2048
Wikipedia	7.16	2.39	1.58	1.58
Code	6.39	2.22	1.82	1.82
Mixed	9.73	3.24	1.68	1.68
Average	7.76	2.62	1.82	1.82

6 Tokenizer Experiments

We conducted 12 experiments across 3 text types (Wikipedia, code, mixed) and 4 vocabulary sizes to inform the 32K production choice.

Key finding: compression saturates at vocabulary 1024 for small corpora, but a 32K vocabulary is necessary for the full physics corpus to capture domain-specific multi-word tokens (e.g., “quantum mechanics”, “wave function”). Inference speed remains stable at ~ 3.9 ms per 100 tokenizations regardless of vocabulary size.

7 Discussion

Contribution framing. This work is a *systems paper*, not a model paper. The contribution is a documented, reproducible pipeline that converts well-known components (BPE, transformers, LoRA, RAG) into a working stack on constrained hardware. We believe this type of work is under-represented in ML literature despite being essential for practitioners.

Who benefits. Three communities: (1) students and independent researchers who need a working baseline without cluster access; (2) applied ML engineers who need operational patterns (checkpointing, retry logic, evaluation harnesses); (3) domain scientists (physics and adjacent fields) who need transparent, auditable foundations for building research assistants.

Limitations.

- The 35.2M model is too small for coherent multi-sentence generation; it serves as a validation vehicle for the pipeline.
- Early-EOS collapse prevents evaluation of generation quality metrics (BLEU, human preference).
- The QA benchmark uses ranking over 4 fixed options, not open-ended generation.
- RAG evaluation uses only 8 queries on a 25-chunk corpus; larger-scale validation is needed.
- All experiments run on a single machine (M3 Pro); multi-GPU scaling is not validated.

Scaling path. The architecture is designed to scale directly: replace TinyLM with a 3B–7B model, expand the corpus to 300M–1B tokens, and add tool-calling agents. The evaluation harness, tokenizer, and reliability mechanisms remain unchanged.

8 Code and Data Availability

All code, trained models, evaluation scripts, and benchmark datasets are publicly available to enable full reproduction:

- **Repository:** <https://github.com/JaydipSingh/slm-physics-assistant>
- **Rust tokenizer:** `src/` — BPE implementation with CLI and library API.
- **Training scripts:** `scripts/` — streaming trainer, LoRA fine-tuning, evaluation harness.
- **Model checkpoints:** Phase 1 base (142 MB) and Phase 2 LoRA adapter (2 MB).
- **Evaluation data:** `data/` — physics QA dataset (20 questions), eval prompts (50), BLEU references, RAG corpus.
- **Results:** `results/` — JSON files for all metrics reported in this paper.
- **Hardware requirement:** Apple M1/M2/M3 with 16 GB RAM, or any CUDA GPU.

The entire pipeline—from corpus download through tokenizer training, model training, LoRA fine-tuning, and full evaluation—can be reproduced in under 6 hours on a single laptop.

9 Conclusion

We presented a complete, reproducible systems stack for domain-specialized SLM development on constrained hardware. Our pipeline—spanning Rust tokenizer, custom transformer with RoPE/SwiGLU/document-packing, LoRA fine-tuning, and multi-metric evaluation—is validated end-to-end on an Apple M3 Pro laptop.

Key results: our 23M-parameter physics-specialized model achieves a QA score of 0.550, outperforming GPT-2 Large (774M) by 69% despite being $33\times$ smaller. We documented and resolved the early-EOS collapse failure mode through systematic architecture improvements (causal masking, EOS-free document packing), providing concrete preventive measures for the community.

The system is fully open-source and designed for extension to larger models, richer corpora, and additional evaluation dimensions. Future work includes: (1) scaling to 3B+ parameters; (2) integration of tool-using agents for multi-step physics reasoning; (3) multimodal extension for physics diagram understanding; and (4) community evaluation on standard benchmarks (SciQ, ARC) to enable broader comparisons.

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